Kakuly Mittal

Mobile No.: +65 8544 0809 | Email: kakulymittal@gmail.com | https://linkedin.com/in/kakuly-mittal

PERSONAL PROFILE

AI/ML Engineer with experience in graph ML, computer vision, NLP, and generative AI, with research spanning GNN-based OOD detection, chip image classification, and target-aware diffusion models. Strong foundation in Python, PyTorch, PyTorch Geometric, SQL, and Power BI, with focus and aim on building production-facing, end-to-end AI systems involving data pipelines, LLM workflows, cloud integration, APIs, and business impact.

EDUCATION

Nanyang Technological University, Singapore Aug 2021 – Aug 2025

Bachelor of Engineering (Honours), Computer Science; Second Major in Business

- Ministry of Education Singapore Tuition Grant Holder
- Specialization in Artificial Intelligence - Honours (Highest Distinction); CGPA: 4.61/5.00
- Coursework: Deep Learning, Computer Vision, Natural Language Processing, Machine Learning, Databases, Artificial Intelligence, Data Science, Data Structures & Algorithms, Software Engineering, Probability & Statistics

WORK | INTERNSHIP EXPERIENCE

Temasek Labs @ NTU – Singapore Sep 2025 – present

Project Officer (Artificial Intelligence & Computer Vision)

- Built and evaluated 10+ GNN-based IP detection pipelines using PyTorch Geometric, including residual GNNs, GINConv, and Graph Transformers; improved F1 score by 7.65% on noisy graph-based chip netlist data.
- Implemented energy-based graph out-of-distribution detection and t-SNE embedding analysis to improve IP localization reliability, achieving a 31.66% relative precision improvement while maintaining >90% accuracy.
- Developed a reproducible computer vision pipeline for chip image classification, curating 240K+ cell images across 3 classes and benchmarking 8 CNN architectures, including EfficientNet, ResNet50, VGG, and U-Net.
- Co-authored and presented two conference papers: accepted paper on GNN-based hardware assurance (**IPFA 2026**) and another accepted paper on chip image classification (**AICAS 2026**).

Chevron Corporation – Singapore Jan 2024 – Jun 2024

Data Scientist Intern, Data Science & Analytics Chapter

- Led end-to-end analysis for an interdepartmental analytics project, integrating 3+ data sources and transforming 10+ tables using Python, Power Query, and DAX to build a Power BI dashboard for business process visibility.
- Enabled cross-functional stakeholders to monitor process performance and identify operational bottlenecks through centralized reporting.
- Built a low-code documentation review application using Power Apps and Power Automate, creating a centralized repository for global teams and projecting a 75% reduction in software documentation review time.

BASF – Singapore Sep 2023 – Nov 2023

Data Analytics Intern (Part-Time)

- Cleaned and transformed 12+ logistics datasets in Power Query and supported development of a Power BI transport management dashboard, improving visibility into logistics spend, shipment trends, and buyer-level reporting.

PSA International Pte Ltd. – Singapore May 2023 – Aug 2023

Data Analytics Intern

- Developed and maintained 2 Power BI dashboards using Power Query, DAX, and Excel, building semantic models and KPIs to support stakeholder decision-making.
- Automated recurring data cleaning workflows using 3+ Python scripts with openpyxl and Power Automate flows, projecting a 50% reduction in manual processing effort and improved data reliability.
- Analyzed container shipment data in Excel to identify demand trends and seasonality, supporting capacity and resource planning for the Shipping department.

KEY PROJECTS

Nanyang Technological University, Singapore Aug 2024 – Present

Final Year Project: Target-Specific Anti-Cancer Peptide Generation using GNNs and Diffusion Models

- Implemented GAT and GCN models using PyTorch Geometric to embed 40K+ peptide-protein complexes, achieving 77% node classification accuracy on unseen data.
- Built a latent cross-attention diffusion model with a U-Net denoising architecture, conditioning peptide generation on binding-site embeddings and achieving a perplexity score of 4.628.
- Co-authored a journal paper based on the project as second author.

Revisiting Kaggle CMI Problematic Internet Use Competition Jan 2025 – May 2025

Team Member (top-5 rank on the Private Leaderboard)

- Developed a robust ML pipeline for predicting problematic internet use in adolescents by integrating advanced data cleaning, feature engineering, and ensemble modelling using XGBoost, LightGBM, and CatBoost.
- Engineered a semi-supervised pseudo-labeling strategy using ensemble regression outputs, improving generalization on a healthcare dataset with over 50% missing values.

AngelHack HackSingapore May 2024 – May 2024

Team Member (3rd Place Winners - Social Responsibility Track)

- Placed 3rd among 80+ teams by developing a responsive community engagement web application for elderly users.
- Built the interactive frontend using ReactJS and Bootstrap, focusing on accessibility, usability, and responsive design.

Text Sentiment Analysis and Domain Adaptation on Customers of Hotels Oct 2023 – Nov 2023

Team Member

- Preprocessed multilingual customer review data using Python libraries including langdetect, contractions, pandas, NLTK, and nlpaug.
- Compared BERT-base and cross-domain BERT review models to evaluate domain adaptation performance on hotel customer reviews.
- Compared BERT-base and DistilBERT to assess the trade-off between model size, performance, and business applicability.

PUBLICATIONS

IEEE 8th International Conference on Artificial Intelligence Circuits and Systems Sep 2026

First author (IPFA 2026 – to be published)

- Topic - Contrastive Graph Convolutional Networks for Hardware Trojan Detection in Third Party IP Cores

33rd IEEE International Symposium on the Physical and Failure Analysis of ICs Jul 2026

First author (AICAS 2026 – to be published)

- Topic - FPGA IP Identification using Graph Neural Network (GNN) with Out-of-Distribution (OOD) Detection

Computers in Biology and Medicine Journal Nov 2025

Co-author (link to the paper)

- Topic - Target-aware latent diffusion model for design of apoptosis-inducing anticancer peptides

CERTIFICATIONS

Amazon Web Services Mar 2026

AWS Certified Cloud Practitioner (Certificate)

- Demonstrates a fundamental understanding of IT services and their uses in the AWS Cloud.

LEADERSHIP EXPERIENCE | CO-CURRICULAR ACTIVITIES

NTU Student’s Union Executive Branch and Legislative Branch (Council) Sep 2023 – Oct 2024

Integration Executive & Council Student Leader

- Chaired a cultural fest with 900+ participants and organized a hall buddy system for international freshmen.
- Led initiatives to improve international student integration, campus belonging, and visibility of multiculturalism.
- Coordinated budgeting and event planning across the Student Integration Committee and six cultural committees to deliver inclusive campus-wide initiatives.

NTU Student’s Union’s Student Integration Committee Sep 2022 – Sep 2023

Vice Chairperson (Student Leader)

- Led the planning and logistics management for two major campus-wide events that focused on cultural education and inclusivity, ensuring successful execution and participant engagement.
- Directed publicity efforts, maintaining high standards to maximize outreach and visibility for events.

TECHNICAL EXPERIENCE

Programming Languages - Python, SQL, C, C++, Java, R

ML/DL: PyTorch, PyTorch Geometric, scikit-learn, TensorFlow/Keras, GNNs, CNNs, Transformers, BERT, RNNs, Diffusion Models, XGBoost, LightGBM, CatBoost

Data & Analytics: Power BI, DAX, Power Query, Power Automate, Tableau, Excel

Cloud & Tools: AWS, GitHub, Azure, Microsoft 365, Figma

Web Development: React.js, HTML, CSS, Bootstrap, Power Apps